Darius Schaub

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Research Profile

I develop machine learning methods to characterize, model, and ultimately control the spatial organization of cells in health and disease. This work has produced the spatial domain identification methods NichePCA and SCALE, and BioContextAI, a community hub for agentic biomedical systems. I now model how multicellular collectives self-organize: MAGNA (accepted at LoG 2026) learns self-organizing Graph Neural Cellular Automata via multi-agent reinforcement learning. My long-term goal is to steer tissues toward healthy states by exploiting their collective intelligence.

Education

Ph.D. candidate in Biomedical AI **Dec. 2022 – exp. Jan. 2027**
University Medical Center Hamburg-Eppendorf (UKE) *Hamburg, Germany*

- Working title: *Characterizing and modeling the spatial organization of cells in health and disease*

Master of Science in Industrial Mathematics **Oct. 2019 – Jun. 2022**
University of Hamburg & Hamburg University of Technology *Hamburg, Germany*

- Grade 1.05 (passed with distinction)
- Thesis: *Solving the traveling salesman problem via deep reinforcement learning*, written in collaboration with Harvard Medical School (supervised by Prof. Yang-Yu Liu)
- Relevant courses: partial differential equations, systems theory, mathematics of neural networks, machine learning, optimization

Bachelor of Science in Nanoscience **Oct. 2017 – Sep. 2019**
University of Hamburg *Hamburg, Germany*

- Grade 1.35 (excellent)
- Thesis: *Ionization dynamics of H_2O – a Matrix Product State study* (supervised by Prof. Daniela Pfannkuche)
- Relevant courses: quantum/statistical/solid-state physics, optimization, numerical mathematics, statistics

Bachelor of Science in Chemistry (switched to Nanoscience) **Oct. 2015 – Sep. 2017**
University of Hamburg, incl. biochemistry and organic chemistry *Hamburg, Germany*

Research Experience

UKE, Center for Biomedical AI **Dec. 2022 – Present**
Doctoral researcher, lab of Prof. Stefan Bonn *Hamburg, Germany*

- Developed two spatial domain identification methods that address the limited scalability and missing multi-scale support of existing approaches: NichePCA [5] and SCALE [2]
- Co-developer and co-maintainer of BioContextAI, a community hub for agentic biomedical systems [3]
- Led and contributed to spatial and single-cell omics analyses of large-scale datasets (Xenium, Visium) in autoimmune kidney disease collaborations [4], [6]–[9]
- Since Nov. 2025: modeling how multicellular behavior gives rise to spatial organization, using Graph Neural Cellular Automata and multi-agent reinforcement learning [1]

Harvard Medical School **Feb. 2021 – Jun. 2023**
Remote visiting student, lab of Prof. Yang-Yu Liu *Boston, MA, USA*

- Channing Division of Network Medicine; focus on deep reinforcement learning, combinatorial optimization, graph neural networks, multi-omics integration and microbiome modeling
- Took part in the VDAART challenge on predicting childhood asthma from multi-omics data [11]

Publications

First and co-first author

[1] **Darius P. Schaub**, Lucia Testa, Robin Khatri, Stefan Bonn, "Learning Graph Neural Cellular Automata via Multi-Agent Reinforcement Learning," in *Learning on Graphs Conference (LoG)*, 2026, (accepted).

[2] Behnam Yousefi*, **Darius P. Schaub***, Robin Khatri, Nico Kaiser, Malte Kuehl, Cedric Ly, Victor G. Puelles, Tobias B. Huber, Immo Prinz, Christian F. Krebs, *et al.*, "SCALE: Unsupervised multiscale domain identification in spatial omics data," *Nucleic Acids Research*, vol. 54, no. 1, gkaf1456, Jan. 13, 2026. DOI: 10.1093/nar/gkaf1456.

[3] Malte Kuehl*, **Darius P. Schaub***, Francesco Carli, Lukas Heumos, Malte Hellmig, Camila Fernández-Zapata, Nico Kaiser, Jonathan Schaul, Anton Kulaga, Nikolay Usanov, *et al.*, "BioContextAI is a community hub for agentic biomedical systems," *Nature Biotechnology*, Nov. 6, 2025. DOI: 10.1038/s41587-025-02900-9.

[4] Cedric Ly*, **Darius P. Schaub***, Robin Khatri, Zeba Sultana, Annika Boxnick, Zheng Song, Tobias Huber, Thorsten Wiech, Eva Tolosa, Ulf Panzer, *et al.*, "Spatial analysis of T cell clonality in autoimmune kidney disease using TRV probes," *bioRxiv*, Sep. 3, 2025. DOI: 10.1101/2025.08.29.673064, (accepted at Cell Reports).

[5] **Darius P. Schaub***, Behnam Yousefi*, Nico Kaiser, Robin Khatri, Victor G. Puelles, Christian F. Krebs, Ulf Panzer, Stefan Bonn, "PCA-based spatial domain identification with state-of-the-art performance," *Bioinformatics*, vol. 41, no. 1, btaf005, Jan. 1, 2025. DOI: 10.1093/bioinformatics/btaf005.

[6] Jonas Engesser*, Robin Khatri*, **Darius P. Schaub***, Yu Zhao, Hans-Joachim Paust, Zeba Sultana, Nariaki Asada, Jan-Hendrik Riedel, Varshi Sivayoganathan, Anett Peters, *et al.*, "Immune profiling-based targeting of pathogenic T cells with ustekinumab in ANCA-associated glomerulonephritis," *Nature Communications*, vol. 15, no. 1, p. 8220, Sep. 19, 2024. DOI: 10.1038/s41467-024-52525-w.

*Equal contribution.

Contributing author

[7] Zeba Sultana, Robin Khatri, Behnam Yousefi, Nikhat Shaikh, Saskia L. Jauch-Speer, **Darius P. Schaub**, Jonas Engesser, Malte Hellmig, Vincent Piegsa, Arthur Hube, *et al.*, "Spatiotemporal interaction of immune and renal cells controls glomerular crescent formation in autoimmune kidney disease," *Nature Immunology*, vol. 26, no. 11, pp. 1977–1988, Nov. 2025. DOI: 10.1038/s41590-025-02291-8.

[8] Huiying Wang, Jonas Engesser, Robin Khatri, **Darius P. Schaub**, Hans-Joachim Paust, Zeba Sultana, Saskia-Larissa Jauch-Speer, Anett Peters, Anna Kaffke, Stefan Bonn, *et al.*, "Type I interferon drives T cell cytotoxicity by upregulation of interferon regulatory factor 7 in autoimmune kidney diseases in mice," *Nature Communications*, vol. 16, no. 1, p. 4686, May 20, 2025. DOI: 10.1038/s41467-025-59819-7.

[9] Junping Yin, Melanie Eichler, **Darius P. Schaub**, Nariaki Asada, Jonas Engesser, Clivia Lisowski, Hans-Joachim Paust, Christina Katharina Weisheit, Jian Li, Daniela Klaus, *et al.*, "Low-dose glucocorticoids attenuate crescentic glomerulonephritis by inhibiting the local differentiation of proinflammatory neutrophils," *Science Translational Medicine*, vol. 17, no. 820, eadu0351, Oct. 15, 2025. DOI: 10.1126/scitranslmed.adu0351.

[10] Sina Abdollahi, **Darius P. Schaub**, Madalena Barroso, Nora C. Laubach, Wiebke Hutwelker, Ulf Panzer, Søren W. Gersting, Stefan Bonn, "A comprehensive comparison of deep learning-based compound-target interaction prediction models to unveil guiding design principles," *Journal of Cheminformatics*, vol. 16, no. 1, p. 118, Oct. 28, 2024. DOI: 10.1186/s13321-024-00913-1.

[11] Xu-Wen Wang, Tong Wang, **Darius P. Schaub**, Can Chen, Zheng Sun, Shanlin Ke, Julian Hecker, Anna Maaser-Hecker, Oana A. Zeleznik, Roman Zeleznik, *et al.*, "Benchmarking omics-based prediction of asthma development in children," *Respiratory Research*, vol. 24, no. 1, p. 63, Feb. 26, 2023. DOI: 10.1186/s12931-023-02368-8.

Awards

Nebius x NVIDIA Build Week Hackathon, Berlin, Germany **Oct. 2026**

- Won the Agentic Deep Research track in a team of three (around 250 participants)
- Built an interactive spatial biology explorer whose agent (custom harness) operates the live viewer and turns raw microscopy images into cited reports via segmentation, clustering, and literature search

MLx Cases Challenge, Oxford ML Summer School, UK **Jun. 2023**

- Won the team challenge among 30 teams: classifying breast carcinoma histology images under limited data

Open Source

spatialdata (scverse) **Mar. 2025 – Mar. 2026**

- Accelerated point filtering by several orders of magnitude (scverse Hackathons in Paris and Berlin)

BioContextAI **May 2025 – Present**

- Co-developer and co-maintainer of the community hub for agentic biomedical systems

Talks and Posters

T Cell Retreat, UKE, Germany **Jun. 2026**

- Talk: *A spatial in situ hybridization approach to T cell clonotype analysis using TRV probes* [4]

scverse Conference 2025, Stanford, USA **Nov. 2025**

- Poster and workshop: *Build your own MCP servers and AI agents with BioContextAI* [3]

ZMNH Retreat, UKE, Germany **Sep. 2025**

- Poster: *A control-theoretic vision for medicine*

Mathematics of Machine Learning 2025, TUHH, Germany **Sep. 2025**

- Poster: *2D tissue reassembly with E(2)-equivariant graph neural cellular automata*

scverse Conference 2024, Munich, Germany **Sep. 2024**

- Poster: NichePCA [5]

T Cell Immunobiology Retreat, UKE, Germany **Jun. 2024**

- Talk: *Spatially informed single-cell resolution analysis of cell-cell interactions*

Teaching and Supervision

Co-led ML-in-practice workshop **Jun. 2025**
University Medical Center Hamburg-Eppendorf *Hamburg, Germany*

- Introduced students to geometric deep learning and Vision Transformers

Teaching Assistant **Sep. 2016 – Jul. 2020**
University of Hamburg *Hamburg, Germany*

- Supervised exercise classes for mathematical courses in the chemistry department

Industry Experience

SICK AG **Jan. 2020 – Nov. 2022**
Data Science working student (incl. internship, Oct. – Dec. 2020) *Hamburg, Germany*

- Reduced scheduled maintenance by 80% by predicting sensor contamination
- Developed and deployed a predictive maintenance and time-series forecasting framework for various sensor types
- Evaluated machine learning algorithms for predicting device errors and maintenance events from limited data

Skills and Interests

Programming	Python (professional), TypeScript (advanced), C/C++ (intermediate), SQL (basic)
Frameworks	PyTorch and PyTorch Geometric (professional), TensorFlow (advanced)
Omics	Spatial and single-cell: spatialdata, scanpy, Xenium, Visium
AI agents	PydanticAI, agent harnesses, tool use, MCP
Technologies	Weights & Biases, Git, CI/CD, Docker, Kubernetes, Postgres
Languages	German (native), English (fluent)
Interests	science fiction, philosophy, politics, dancing, fitness